

WASP-49b

R-1226

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-49_200106 · created 2026-07-31T09:19:50+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2458854.7045 +/- 0.0031 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.135 +/- 0.041
Transit depth (Rp/Rs)^2	1.82 +/- 1.1 [%]
Orbital Inclination (inc)	83.83 +/- 2.86
Airmass coefficient 1 (a1)	0.81 +/- 0.43
Airmass coefficient 2 (a2)	0.06 +/- 0.18
Scatter in the residuals of the lightcurve fit is	48.34 %
Best Comparison Star	#1 - [34, 244]
Optimal Aperture	1.74
Optimal Annulus	7.97
Transit Duration (day)	0.07 +/- 0.043

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.135 ± 0.041 vs 0.116 (deviation 0.46σ)
Duration fit vs published	0.07 d vs 0.09 d (deviation 0.47σ)
Mid-time O–C	7.2 min
Depth SNR	3.3
Residual scatter	48.34 %

Light curve

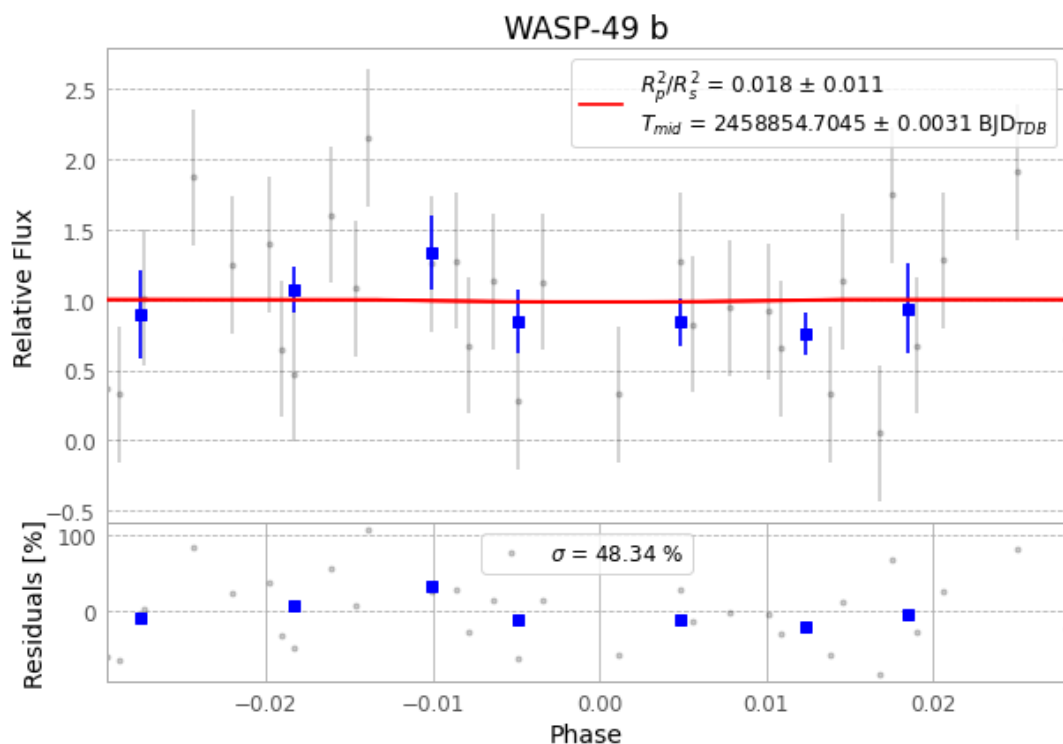


Figure 1 — FinalLightCurve_WASP-49 b_2020-01-05.png (SHA-256 71e9ee529b7c1104...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [256, 176], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d4df735accf679e8...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

83 FITS frames (55 MB), WASP-49200106024812.FITS → WASP-49200106065412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-49-200106.log (90f7450755c8...), FinalLightCurve_WASP-49 b_2020-01-05.png (71e9ee529b7c...), FinalLightCurve_WASP-49 b_2020-01-05.csv (3bfef697d565...)

Compute resources

Wall time 140.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 09:19Z.